AI Portfolio – Autonomous Smart Cleaning Agent

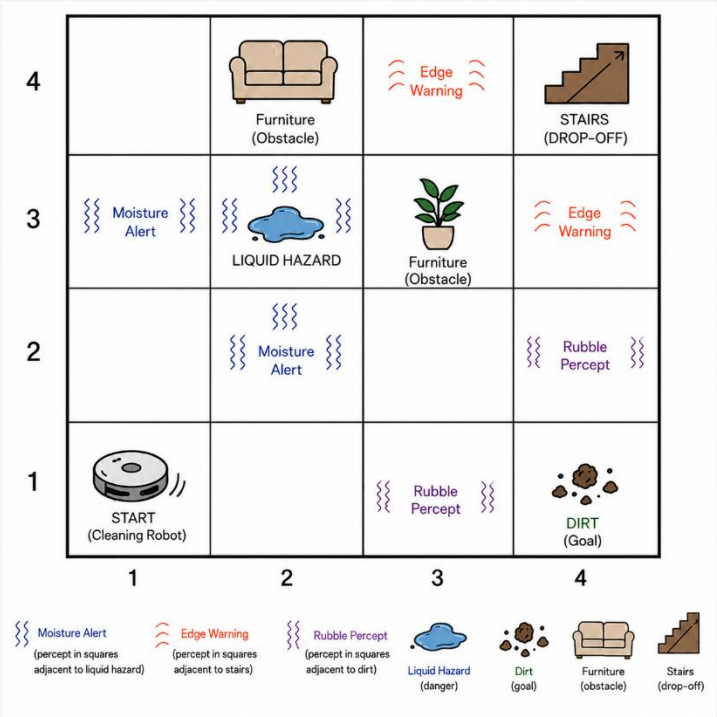
Problem analysis & AI system design

For this assignment I have chosen to do domain 1 (the Wumpus World), adapted to the real-world representation of an autonomous smart cleaning agent. This agent is designed as a robotic vacuum cleaner operating on a 4x4 grid-based environment, that represents a floor space such as a carpet. Its main goal is to look for dirt/mess to clean up whilst avoiding the main threat of a liquid hazard zone.

The main hazard is the liquid spill zone that could damage the robot. The environment also includes edge warnings, to prevent the agent from entering unsafe areas represented as stairs. Lastly, my twist is the addition of blocked cells representing furniture. I used furniture as the twist as I believe it's a realistic part of the environment and will help demonstrate clearly how my agent could work in real-world households.

The agent moves through this environment, updating its knowledge base based on nearby percept's and will use inference rules to determine the safest movements. A visual representation of my environment is at appendix A below.

APPENDIX A *(AI image, Generated by ChatGPT)*



Problem analysis

The robot's objective is to navigate the environment efficiently, avoiding hazards and cleaning all dirty areas. However, the environment is partially observable, so the agent can't view the whole grid at once, as a result, it may not initially choose the most efficient route.

Furthermore, the liquid hazard is hidden from the agent, until it comes into a square adjacent to it. This means the agent may unknowingly move towards a dangerous area. The blocked cells will also restrict movement, forcing the robot to find an alternative route.

The environment is static because the layout and hazards do not change while the robot operates. It is also deterministic, as the same action always produces the same outcome. The system uses discrete movement because the robot moves between fixed grid cells.

Justification of approach

My AI approach is suitable as my environment has hidden information for the agent to discover. It also includes hazards that the agent can infer, learn from, and make decisions based of the knowledge it's found. Furthermore, my project uses a small environment that works well for agents with deterministic characteristics.

My chosen environment also helps explain why alternative AI approaches would be less suitable, as if I used a model that requires large amounts of data and training such as machine learning, it would be unnecessary for an environment of this size and scale.

Algorithmic plan

Component	**Description**
Liquid hazard	Adjacent squares contain moisture percepts. The hazard square is unsafe
Furniture (obstacles)	Cannot be percept. Blocks paths; an agent cannot go through this square
Stairs	Adjacent squares contain edge warning percepts. Stair squares are unsafe.
Dirt (goal)	Adjacent squares contain rubble percepts. When the agent reaches the dirt square, it cleans the dirt and completes the task.

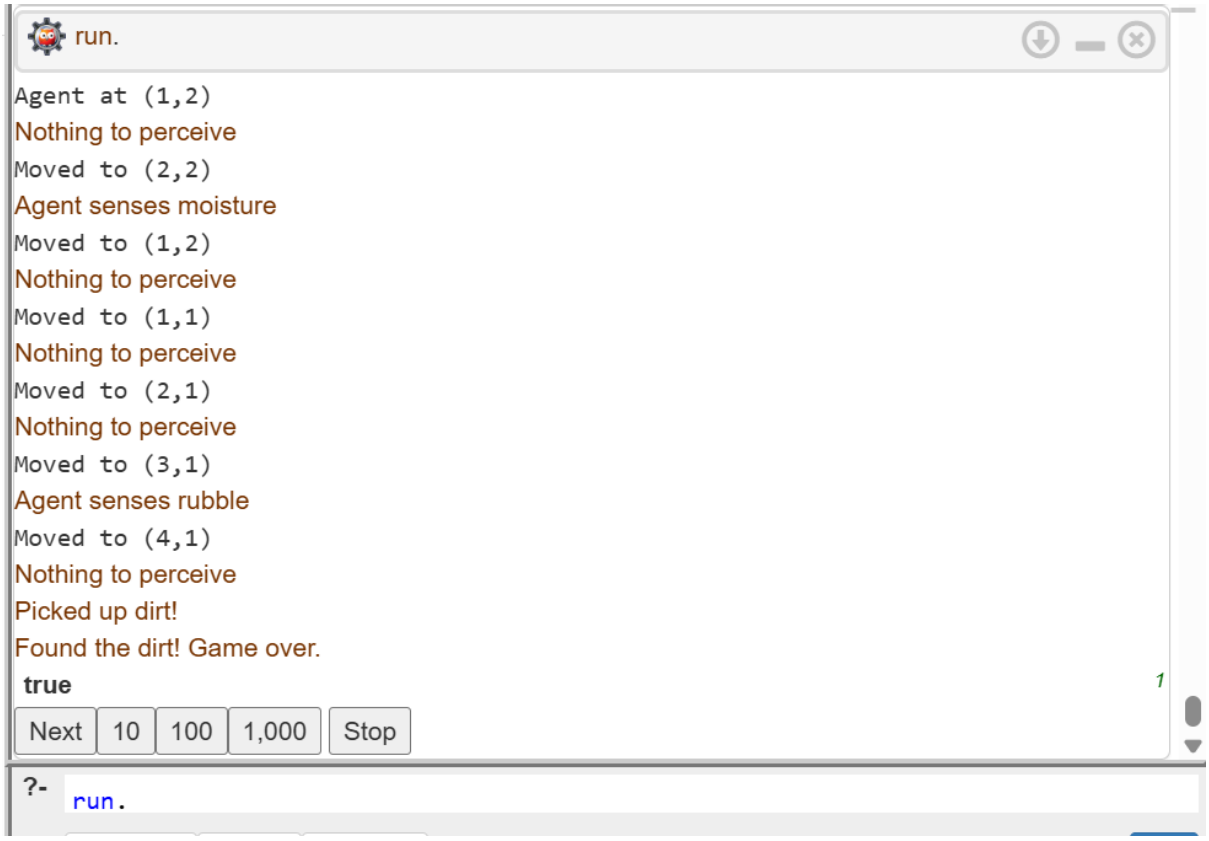
Programme rules:

- A square is safe if it has no moisture or edge warning.

- An agent can move to a square if it is adjacent to the agent and it is classified as safe
- An agent moving to an unsafe square result in failure
- If an agent perceives rubble, it should transition to goal state and win the game

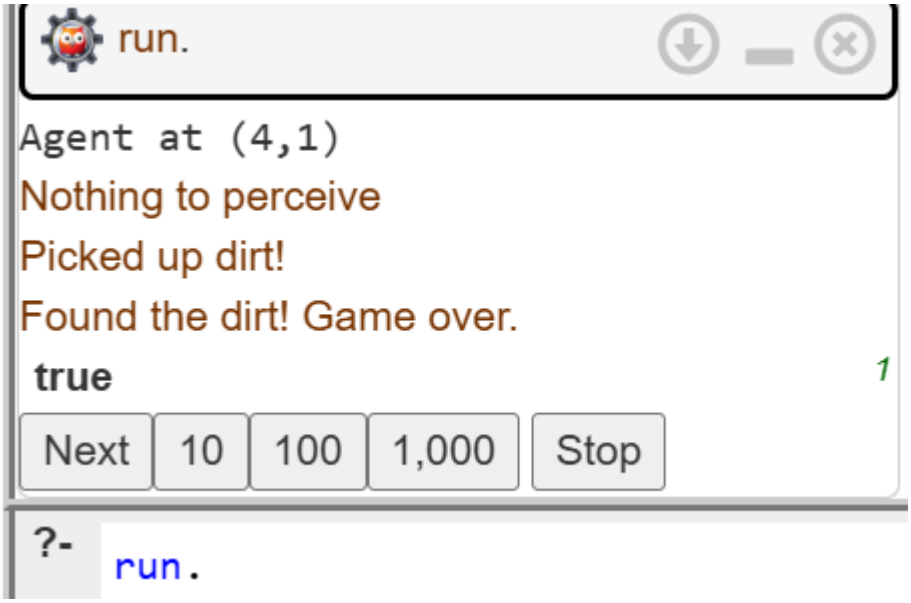
Implementation

Test 1 (Running from position [1,2])



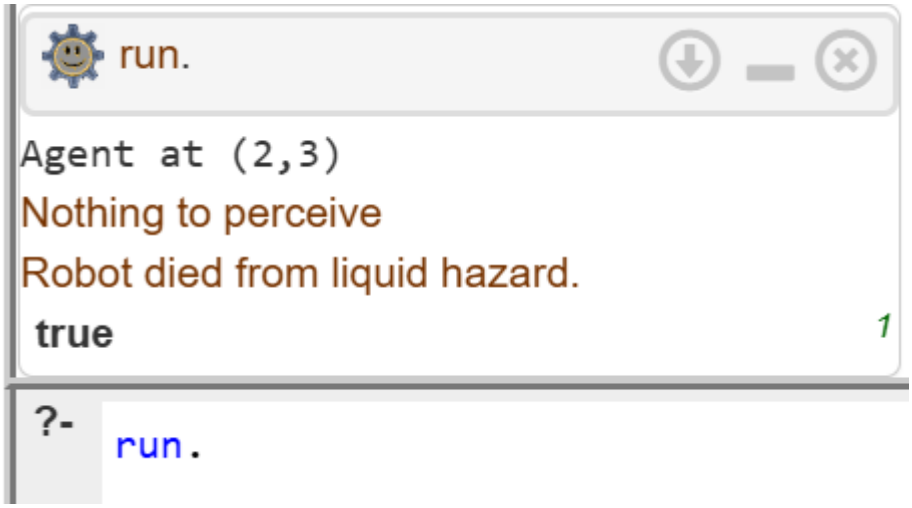
The agent moves past the liquid and near the dirt during this run, meaning it should percept moisture at (2,2), and rubble at (3,1) during its route, which it does. Once the agent finds the dirt, it picks it up and finishes the program.

Test 2 (Running from edge case: position [4,1] (dirt square))



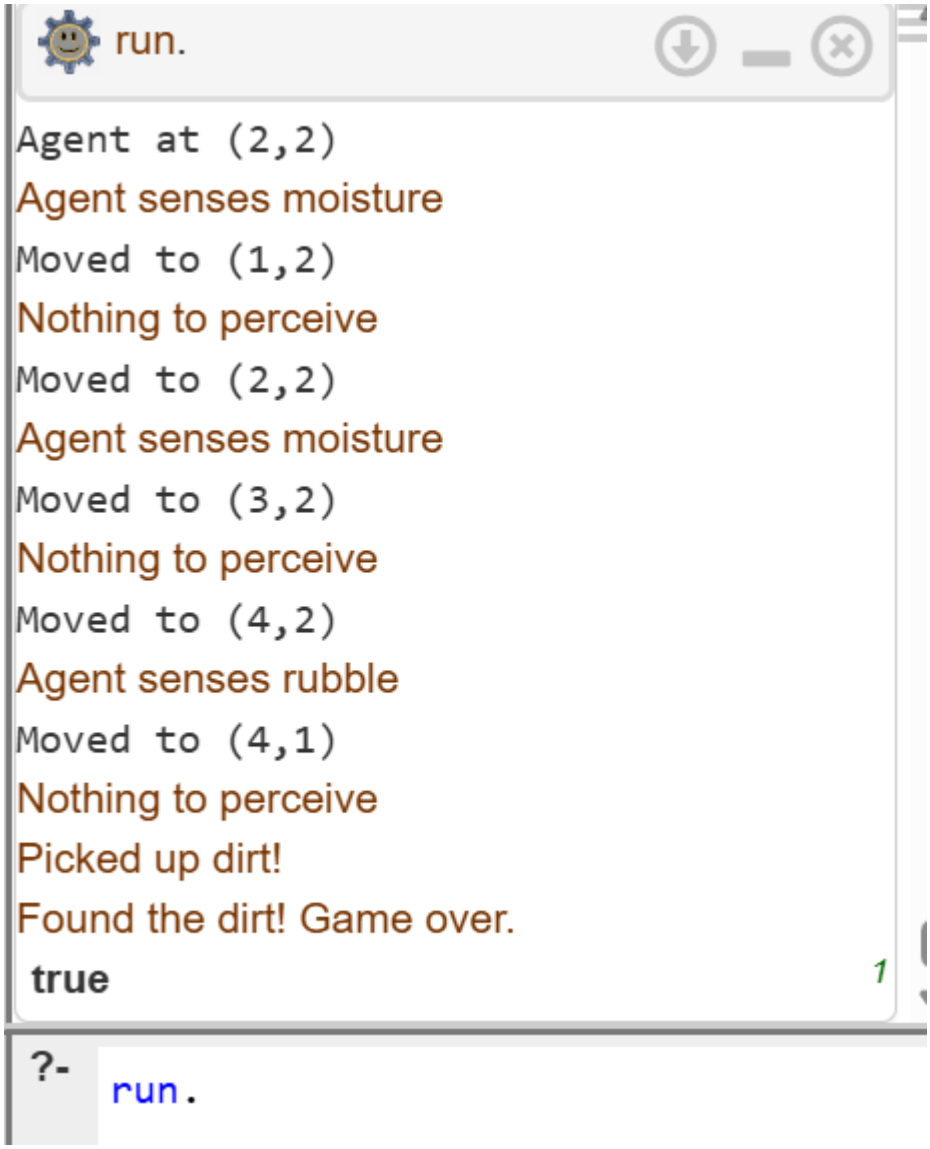
I tested what would happen if the agent started on the goal square, and it acted correctly by instantly picking up the dirt and finishing the game.

Test 3 (Running from edge case: position [2,3] (liquid hazard square))



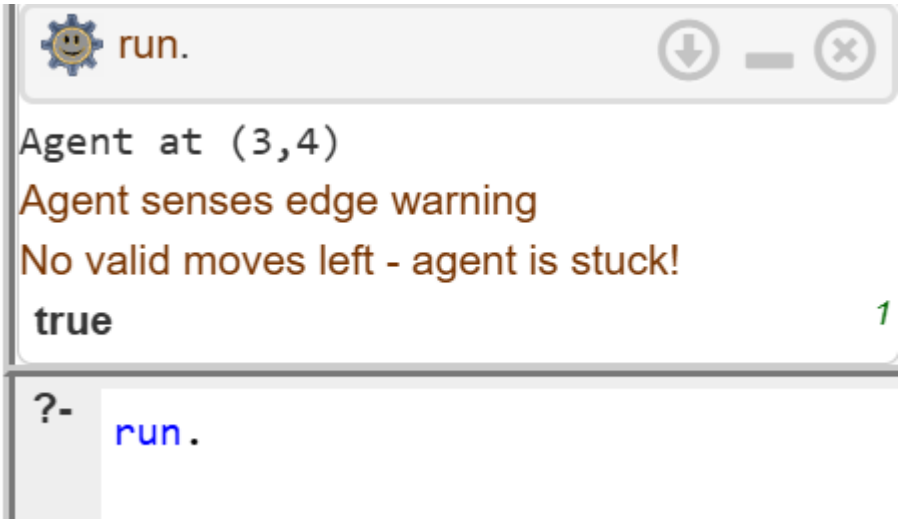
This tested what would happen if the robot started on the hazard square that should be game ending. My program ran as expected and ended the game instantly.

Test 4 (Running from position [2,2])



Basic test from a different starting position. The agent navigates, perceives, and wins as expected.

Test 5 (Edge case: position [3,4])



In this test the agent is placed in a position that is blocked with furniture. The result is the expected output.

Reflection, and Ethical Analysis

Introduction

My system overall is an autonomous cleaning agent that uses rule-based AI and deterministic reasoning to navigate the 4x4 environment to complete its goal. It uses percept-based interface, and the agent needs to explore the environment to help understand it is partially observable. During my reflection, I will compare my system with advanced systems that use modern deep learning techniques.

Advanced AI Contrast

An advanced AI system could be a learned agent rather than a rule based one. In this example system, decisions would be black-box and be generated through probabilistic reasoning rather than exact symbolic rules. This means the agent would be able to reason under uncertainty and incomplete information, allowing the agent to adapt effectively to different and new environments. Also, unlike a symbolic system that relies on percepts to identify parts of the environment, an advanced AI system could infer the environment directly from visual sensor data. Furthermore, a modern learning agent would be able to learn and optimise routes automatically, understanding the environment and improving performance over time. For this domain specifically, recognising furniture and identifying hazards to optimise the best cleaning routes.

Despite this, advanced systems will have limitations for this problem domain. The advanced models may behave unpredictably and make decisions that are harder to understand or trace. This black-box nature causes issues as it makes it harder to maintain and debug when mistakes happen, as well as bringing up safety concerns as

the agent could act in dangerous ways without any clear reason. A clear advantage of a symbolic approach is its transparency and explainability; every decision the agent makes can be trace and explained by rules and percepts. In contrast, advanced learning-based systems are harder to trace as the decisions made are without clear reasoning, making it much harder to debug, understand, and trust the system. Moreover, another difference is how learning models require large datasets to work, using up much more space and computational power than a rule-based system that requires no training.

Ethical Reflection and Trade-Offs

One ethical point that can be made for robot vacuums is how it would impact the user's privacy. For example, advanced systems will learn room layouts and map homes in its attempt to find the most optimal route. This would require the system to collect and store user data, which is a major ethical issue. It asks questions such as if the user gave consent for this to happen and is aware of where the data is being held or what else it is being used for.

Moreover, modern systems that use learn-based reasoning can reason under incomplete information, making it potentially dangerous for a user to be over-reliant on the agent when it could not be as trustworthy as the user may believe. For example, if the robot makes an error and misses parts that need to be cleaned it could leave dirt or other hazards that could cause damage to property or potential safety risks depending on if the owner has children/pets. This is a trade-off for this type of system but also turns into an ethical concern if people stop paying attention to the environment and become too reliant on the autonomous system, affecting the safety of the house/area.

Another quite relevant point is accountability; who is responsible for the actions/outputs of the robot. For example, if the robot damages property due to failure to complete its task, or even an error leading to destruction if it followed an incorrect path, it's unclear who takes responsibility for its actions. Responsibility could potentially fall on the manufacturing company, software developers that worked on the system, or maybe even the user employing the system. Using an advanced learning-based system can complicate this even more due to its black-box nature and how decisions are not explicit rules but complex learned behaviour. This can cause issues over liability as it would be unclear due to there being many people and variables involved in the robot, leading to a lack of transparency and frustrating accountability issues.

Lastly, an interesting trade-off is between control/predictability and capability. Symbolic systems have less capability as they are limited to the chosen rules and have no training or learning capabilities; but they are a lot more controllable and predictable in the way the agent will work. This can benefit testing and debugging, as the system is

easier to understand and trace. On the other hand, advanced models have huge potential and capabilities as they learn and improve in efficiency with every use; but they can be very unpredictable by nature, making the agent hard to control and understand. Although it can be useful to use a system that has more capabilities to optimise performance; in physical environments, like the one a robot vacuum would operate in, unpredictability can become a dangerous characteristic and create safety concerns around property, people, or even pets.

Conclusion

Overall, I believe my symbolic rule-based system is appropriate for this problem domain as it works well with smaller environment and is easier to understand and maintain due to its predictable and explainable nature. On the other hand, advanced AI systems that use a learning-based model are much more adaptive to all environment types and have the capability to perform the same task in an optimised and efficient way which can be necessary for larger environments/problems. Nevertheless, increased capability introduces a much higher number of ethical trade-offs as the system becomes smarter but uses more data and is harder to predict, control, and understand. In conclusion more intelligent systems are not automatically safer or more ethical but can be highly efficient in appropriate environments.

Generative AI tools declaration

Generative AI tools (ChatGPT) were used during the development of this portfolio only to support planning and proofreading. All final content, design decisions, and implementation was reviewed, wrote, and understood by me.

Generative AI tools (Claude) were used during code development to help me design the movement for my agent as this was not discussed during lectures/labs. I created my own movement loop at first but then ran into an issue where my agent would get stuck in a loop between two squares. After using Claude to help me understand what the bug was, I now know that I needed to add a visited list of squares to the movement loop in order to stop the agent immediately going back the way it came; by adding the destination square (NX,NY) to the list instead of the origin square (AX,AY) the agent was able to navigate the environment with freedom. Other than this support, my code was fully written and designed by me.

My use of generative AI was in accordance with the modules permitted AI use policy.